Journal of Educational Change (2023) 24:479–499
https://doi.org/10.1007/s10833-022-09453-7

How can education systems improve? A systematic literature review

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Accepted: 3 March 2022 / Published online: 7 April 2022
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Abstract

Understanding what contributes to improving a system will help us tackle the problems in education systems that usually fail disproportionately in providing quality education for all, especially for the most disadvantage sectors of the population. This paper presents the results of a qualitative systematic literature review aimed at providing a comprehensive overview of what education research can say about the factors that promote education systems' improvement. This literature is emerging as a topic of empirical research that merges comparative education and school effectiveness studies as standardized assessments make it possible to compare results across systems and time. To examine and synthesize the papers included in this review we followed a thematic analysis approach. We identify, analyze, and report patterns in the papers included in this systematic review. From the coding process, four drivers for system improvement emerged: (1) system-wide approaches; (2) human capital; (3) governance and macro–micro level bridges; and (4) availability of resources.

Keywords Educational change · System-wide improvement · Comparative education · International education

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Introduction

The appearance and consolidation of national and international standardized assessments resulted in an epistemological revolution in comparative education. We now know -to a certain degree- which education systems improve over time in terms of learning outcomes. This measurability shift contributed to the emergence of multiple studies that focus on how education systems improve. Through a systematic review of the literature, this paper aims to understand the efforts made through educational research to explain and interpret education systems' improvement. Our research question is: How does educational research explain the improvement of educational systems? To address this question, we offer a systematic literature review that considers studies across methodological approaches and contexts.

This paper is a qualitative systematic literature review of empirical research on the topic of systems´ improvement. We analyzed and synthetized a particular body of literature to compare theoretical, methodological, and empirical findings (Snyder, 2019). We used a thematic synthesis approach (Booth et al., 2012) to provide a comprehensive overview of the topic of education systems' improvement, and to identify common patterns in the literature in terms of the type of interventions that could explain systemic improvement.

Although there are different definitions in the literature of what constitutes an improvement in an education system, Levin offers a definition that is useful to clarify our focus: "Bringing more students than ever before to higher levels of achievement than ever before, on a broader range of skills and attributes than ever before, with less inequity in outcomes than ever before" (Levin, 2012). In this article, we will use this broad definition closely related to the possibilities of measuring learning outcomes.

There has been considerable writing about the best performers in international large-scale assessments (ILSAs) in the last decades. This body of research reveals the characteristics that explain these educational systems' success (Barber & Mourshed, 2007; Jessop, 1998; OECD, 2019; Rivas, 2015; Rivas & Scasso, 2021; Sahlberg, 2011; Schleicher, 2012, 2018; Sellar & Lingard, 2013; Steward, 2012; Tan, 2019; Tucker, 2011). Studies on factors associated with learning outcomes came along the same organizations that applied standardized assessments (Alves, 2008; Crato, 2021; Gómez & Suarez, 2020; LLECE-UNESCO, 2013; OECD, 2019). Others used econometric methods to find the possible explanations for the success of the systems with better results (Rindermann & Ceci, 2009).

This literature is relevant, but it mainly aims to obtain a static image of successful education systems, a photograph of the features that contribute to top student learning results. Our interest in this paper is different. We aim at understanding processes of improvement and the processes of change involved in improvement.

The theme of positive change has been thoroughly developed in the fast-growing field of school improvement. There are journals specifically dedicated to educational improvement at the school level,[1] academic societies,[2] and international conferences.[3] However, as we will later show, the theme of education systems' improvement has received less attention and is more fragmented.

Although the efforts to understand how education systems improvement can nurture from the school-system improvement literature (e.g., David & Talbert, 2012; McAleavy & Elwick, 2016; O'Day & Quick, 2009), there are considerable differences in designing a research study to analyze change in an education system vis a vis a limited group of schools.

There is a need to understand better why and how education systems improve. As Snipes et al. (2002) suggested, we need to discuss what empirical studies can tell us about the kind of interventions that can strengthen education systems. Understanding what contributes to improving a system will also help us tackle the problems in education systems that usually fail disproportionately in providing quality education for all, especially for the most disadvantage sectors of the population. However, it is also relevant to develop a critical, contextualized, historical, and reflective approach to systems improvement. As we will show in this paper, many studies fall on the promise of recipes and oversimplification of educational systems through the limited gaze of ILSA. Our aim is to both consider the relevance and the epistemological challenges of studying education systems' improvement.

In this paper, we attempt to analyze publications on systemic improvement in education by looking at (a) the criteria that are used to define that a system has improved; (b) the research approaches that are used to study systemic education improvement; and (c) the types of explanations that are offered to establish that certain factors contribute to systemic improvement.

We will start by showing the critical links between these empirical studies and some traditions and theories on systems` improvement to frame our analysis. We will move from the first comparative travelers to the development of *Program for International Student Assessment (PISA)*, the publication of the McKinsey Reports that refer to systems' improvement, and the propagation of RCTs designs in educational research. We then present the methodological strategies we used to identify a series of empirical papers and reports that have tried to understand how education systems improve over time. Finally, we examine the texts that we identified by considering the methodological approaches used, the main conclusions drawn from these studies, and their contribution to the literature on systems` improvement.

[1] For example, Improving schools https://journals.sagepub.com/aims-scope/IMP.

[2] School effectiveness and school improvement https://www.tandfonline.com/action/journalInformation?show=aimsScope&journalCode=nses20.

[3] For example, International Congress for School Effectiveness and Improvement https://www.icsei.net/about-icsei/.

System-wide improvement in the education literature: A brief view over time

The origins of research that compares the quality of education systems can be traced to the writings of intellectuals and administrators that travelled around the world searching for models to imitate. The analyses of these travelers (Noah & Eckstein, 1969) and the definition of the best systems were based on direct unsystematic observation of organizational models and pedagogic practices. Transfer of models happened from one education system to another, with some systems, such as Prussia and France, becoming the primary loci of attraction. After the Second World War, with the advent of notions of development, international organizations became key actors in identifying and promoting best practices in education policies (Beech & Lista, 2012; Beech & Rizvi, 2017).

During these pre-ILSA phase comparisons tended to be spatial (among different systems) more than temporal (the same system across time). The aim was not to find systems that improved but those that were better than others. Some indicators that measured access and student trajectories were used to define the quality of a system. However, overall comparisons were based on exogenous factors to the education system (economic performance or technological development) partly attributed to education quality. For example, the education system of the Soviet Union attracted the attention of US researchers after the launch of the Sputnik satellite (Trace, 1961). Similarly, the economic boom of the "Asian Tigers" captivated education specialists looking for models to imitate (Morris, 1996). The causal link between economic or technological development and education success was relatively weak.

The creation of standardized assessments revolutionized the possibilities of comparing education systems (Addey et al., 2017; Arnove, 2015; Steiner-Khamsi, 2016; Verger et al., 2012). National standardized assessments allowed for comparisons between subnational education systems and a new research focus emerged, especially in the United States, with the possibility of comparing multiple districts of education (Elmore & Burney, 1998; Murphy & Hallinger, 1988; Snipes et al., 2002; Wenger et al., 2002).

With the advent of ILSA, cross-national measures of student achievement started to be accepted in educational research as an objective indicator of the quality of an education system, validating intra-system comparisons and rankings.

PISA became the leader of the measurement shift. As a “universal” indicator of education quality, PISA had several effects on the literature on system-wide improvement in education. First, it enabled a more robust spatial comparison, and through its rankings became a Global Positioning System in education (Novoa & Yariv-Mashal, 2014). Second, PISA’s rankings contributed to the visibility of the “world champions" of education. The best education systems are now defined through advanced psychometric techniques in the hands of an allegedly neutral external observer. Third, ILSA, and PISA in particular, allowed for temporal comparison. By measuring the quality of learning outcomes through time, it becomes possible to identify the systems that have made the most significant improvements and not only the best in absolute terms. Even though the comparability of tests such as PISA across time have been debated (Fernández-Cano, 2016; Hopfenbeck et al., 2018; Lindblad et al., 2015; Rivas et al., 2020), the use of these tests to identify significant improvements

shifted policymakers' and experts' attention toward systems that had received little attention in the past, such as Estonia or Portugal (Boman, 2020; Carvalho et al., 2017; Mikk, 2015). On the other hand, the publications on system-wide improvement have had an impact, not only within the academic field but also with actors such as international organizations and global for-profit consultancy firms.

This brief overview shows that the theme of system-wide improvement is relatively new and had a great impact with the development of ILSAs (mainly PISA). However, the emerging literature on systems improvements is related to different established epistemologies and theoretical traditions.

Conflicting theories on systems' improvement

One of the most influential theoretical approaches to systemic improvement in education is rational choice theory. This theory promotes the idea that improvement in education can and should be attained by modifying actors' incentives. The different variations in the rational choice narrative are based on the notion that individual agents shape social structures as they make rational decisions to maximize their benefits. Thus, studies within this approach aim to find the factors that determine individual choices and are usually associated with human capital theory (Becker, 1976) that sees education as a means to maximize economic performance. In its applied version, rational choice is associated with New Public Management that promotes the decentralization and de-bureaucratization of the public sector, combined with tight central control of results in standardized exams, fiscal discipline, and an emphasis on the management styles of the private sector (Hood, 1991, Osborne & Gaebler 1992).

These perspectives are construed as evidence-based and tend to reject the possibility of questioning the ideological and theoretical conceptions that sustain their findings. For example, Hanushek and Woseman (2007) present their work as providing access to a universal truth that applies to every context independent of each educational system's historical, economic, cultural, and political conditions.

One of the most noteworthy examples of this kind of approach is Barber et al. (2011) "Deliverology 101: A Field Guide For Educational Leaders". The book promotes reforms based on pressures for performance in standardized tests and has been very influential with policymakers looking for recipes to improve educational systems worldwide.

This narrative is also linked to the approach based on randomized controlled trials (RCT) in education (Dhaliwal et al., 2013; Masino y Niño-Zarazúa, 2016; McEwan, 2015; Murnane and Ganimian, 2014; Snilstveit et al. 2016). This methodology has become sanctified with an aura of scientific authority as it became the gold standard to evidence causality in educational interventions. The narrative based on RCTs, even with its promise of objectivity, faces at least two problems. The first problem is related to the transfer of policies from one context to another (Edwards, 2018). RCTs can establish causality—even though this can be debatable (Deaton, 2020). However, the impossibility to replicate precisely the same intervention in another context annuls potential political effects of findings based on RCTs. The second

problem of this master narrative is that in its search for scientific rigor, it focuses on interventions that are methodologically isolated from their context to make them measurable.

Every thesis has an antithesis. In contrast with the performance–based reforms and the positivistic methodological bias, the cultural approach to educational change understands educational improvement as a complex process that involves different actors, identities, histories, and tensions that are an essential part of every education system. One of the most paradigmatic authors in this line of thought is Torsten Hussen (1994), who some decades ago pointed to the complexity of educational reform by highlighting a series of guiding principles: (a) educational reforms should always be part of more comprehensive social reforms; (b) reforms should be perceived as a need from within the educational system; (c) educational reforms are inherently slow; and (d) the broader the scope of reform, the greater the need for wide-ranging consensus.

Following this logic, Elmore highlights the strategic importance of addressing complexity. He notes that systemic problems require interventions that are multidimensional, integrated, highly focalized, coherent, and based on an explicit theory of action (Elmore, 2007). This perspective was further elaborated by Fullan (2016) when promoting his Whole System Reform. Overall, these approaches rest upon historical analyses of the grammar of schooling and the invisible and permanent organizational principles that structure school systems and tend to be resistant to reform efforts (Tyack & Cuban, 1995).

Cultural approaches to education reform combine the fields of political sciences with organizational theories and the analysis of pedagogic dimensions. In this sense, they are more comprehensive and attuned to complexity than theories based purely on a quantitative focus. However, they can also be seen as an idealistic perspective that does not account for most educational systems' actual material and political contexts. They address variables such as culture and the need for broad consensus that is very difficult to measure and change.

Another relevant perspective that is not as common in the literature on education policy is the analysis of curricular governance within educational systems (Cohen & Spillane, 1992; Schimdt & Prawat, 2006). This approach aims to overcome the simplicity of economic perspectives and the abstract construction of cultural theories. Authors within this perspective suggest that to understand how learning can be improved at a systemic level, it is necessary to focus on the curriculum as a critical determinant of students' learning by asking the following questions: How is the curriculum defined and structured? Through which artifacts does it get into classrooms? How is it transformed and reinterpreted in teaching and learning practices?

The book *"Why Schools Matter: A Cross-National Comparison of Curriculum and Learning"* (Cogan et al., 2001) is the archetype of this approach. This extensive research project mapped the comparative curricular trajectory in 54 educational systems based on data from TIMMS. The curricular policy was divided into four elements: learning aims, contents, teaching methods, and assessment forms. The trajectory from the prescribed curriculum to actual learning was studied comparatively. One of the main conclusions of this study is that education systems with a centralized curriculum tend to attain more consistency in students' learning, especially if

they can successfully align the different phases of curricular flow. The authors suggest that textbooks can be essential in systematically coordinating student learning across a system. There needs to be a 'live' institutional center that ensures the coherence of the curriculum (Cogan et al., 2001). The State agencies in charge of institutionalizing the curricular design, trying to regulate the learning trajectories with constant feedback based on practices, are the neural centers in education systems with the best results.

Another perspective became widely famous in recent years. The release of the report "*How the world's most improved school systems keep getting better*" by the global management consulting company McKinsey & Company in 2010 was a significant turning point in the literature on systemic improvement in education (Mourshed et al., 2010). This was the second major report that McKinsey & Company published. The first report, "*How the World's Best-Performing School Systems Come Out On-Top*," was published in 2007. From the first to the second report, the attention shifted from studying the world champions of standardized tests to the champions of improvement in education. The publication had such an impact that it became one of the most cited and influential reports in international educational policies (Coffield, 2012).

Based on 200 interviews, the authors systematized 575 education policy interventions in the 20 education systems that they analyzed. The findings show that these systems follow different paths toward improvement but that these paths are remarkably similar depending on their point of departure in the global league table of educational performance. The "journey" of these systems toward improvement illustrates inverse trajectories, depending on their starting point: High prescription and regulation for those in the lower echelons of learning performance, moving toward progressive professional development for teachers for those in the middle, and then a shift toward devolution of pedagogical responsibilities for decision making to teachers for higher-performing systems. Nevertheless, in the end, the Report suggests that there is a unique linear recipe for educational systems for each of the predefined starting points. Cultures and contexts play minor roles, as long as policymakers *get the fundamental right*.

The second McKinsey Report invites policymakers around the world to follow a linear and straightforward instructions manual; they should locate their education system in the league table of learning performance and depending on whether they are among the "poor," "regular," "good" or "very good" performing systems, follow the recipes that the Report indicates for their level and in that way, they would attain improvement.

This perspective was the object of much adverse criticism. Some criticized the lack of attention to the complexity of causality in the Report, which converts a series of conjectures and unfalsifiable hypotheses into recommendations for action through a process that would not stand a test of scientific validity (Morris, 2015). Others centered their critique on the lack of a cultural, political, and historical sensibility. The McKinsey Report compares educational systems with very different historical traditions and trajectories without considering their context. The text falls into a soft comparison that misses the localized complexity of improvement processes in each

education system (Auld & Morris, 2016; Coffield & Williamson, 2011; Feniger & Lefstein, 2014).

Even with its weaknesses and the critiques it received, the McKinsey Report was an important addition and, in many ways, a catalyst for the growing number of studies that are aimed at understanding the improvement of education systems. The literature's typical features within this thematic focus are not so much their theoretical frameworks or varied methodological approaches but rather their quest for explaining why and how specific education systems improve over time. This same interest we pursue in this article through a review of the available literature.

Methodology

The clash of theoretical approaches set the context to study systems' improvement. In this section we present the methodological framework of the systematic literature review. Our focus was on research aimed at explaining education systems' improvement. As we aimed at understanding common themes, methodologies and theoretical perspectives we used a thematic synthesis approach (Booth et al., 2012).

To select the papers we included in this review, we followed four main strategies. Firstly, we completed an internet search in Google Scholar using the search terms "Education" AND "System-wide Improvement" AND "high schools" AND "Elementary" AND "Education Reform." That initial search yielded 1230 results. Secondly, we identified prominent researchers working on system-wide improvement and searched text from these authors. Thirdly, we checked the reference list in all the relevant papers, either written by renowned scholars or publications having more than 100 citations on Google Scholar, to identify potentially relevant texts for our review. Lastly, given the importance of PISA tests, we also did a country-based search to identify the top ten countries that had improved their PISA scores significantly between 2003 and 2018.[4] Once we identified these countries, we specifically looked for papers that tried to explain these scores' improvement.

The combination of these four strategies provided 1970 results. One member of the research team read the title and the abstract of all 1970 results. These papers were classified into four different categories: (1) comparative papers about system improvement; (2) comparative papers about improvement at the subnational level; (3) papers about an individual system whose research question or research purpose explicitly referred to the notion of system-wide improvement; and (4) papers about an individual system whose research question or research purpose was ampler and not explicitly focused on the notion of system-wide improvement. Out of the 1970 articles, 328 fell in one of the first three categories. Those were the papers we decided to continue working with.

Once the 328 articles were identified, two members of the research team read the full papers. We reduced the number to 33 papers by removing papers that did not

[4] Our search countries were Albania, Qatar, Estonia, Portugal, Poland, Peru, Ireland, Russia, Israel, and Slovenia.

focus explicitly on system-wide improvement as their central theme (See Appendix A for a list of the papers included in this review). Papers that were tangentially about improvement but were not considered in the review fell under one of the following categories:

1. *Not focusing on improvement* These papers analyzed reforms in a given system that had improved in certain areas but tended to have a critical view of the reforms and how improvements were considered. They mainly questioned the idea that those systems had improved (e.g., Nguyen et al. 2013; Paterson, 2019).
2. *Not identifying system-wide improvement* These papers mentioned positive aspects of an education system or a reform, such as improving school leadership skills but did not identify a system-wide improvement (e.g., Graczewski et al., 2009; Hallinger & Heck, 2011).
3. *Not explaining 'why' a system improved* We also decided to discard papers that did not examine why a system improved (even within a loose notion of causality). Some of the papers we reviewed described reforms and then mentioned that a system improved but failed to bridge one thing with the other (e.g., Dykstra, 2006; Hopkins et al., 2007).
4. *Mostly describing and systematizing education reforms* Some of the papers that we had first identified provided comprehensive descriptions of reform. Notwithstanding, the papers' purpose was not to offer a consistent analysis of the education system's results (e.g., Lapping, 2004; Sam & Riggan, 2013).
5. *Analyzing particular micro-level reforms* We also excluded papers that only looked at a particular reform or improvement in one specific area. Rather than addressing system-wide initiatives, these papers focused on specific dimensions of the education system, such as literacy instruction (Quick et al., 2009) or reading skills (Betts et al., 2005).

To examine and synthesize the papers included in this review we followed a thematic analysis approach. We tried to identify, analyze, and report patterns in the papers included in this systematic review. We borrowed from Glaser and Strauss's (1967) coding approach for grounded theory to organize our thematic analysis. Therefore, the corpus was analyzed by an interpretive process that moved from concrete to more abstract levels using a three-tiered process. The process moved from (a) assignment of individual codes to portions of data that carried discrete meaning, to (b) grouping of these codes into "axial" codes, and then (c) to statements of thematic convergence across the studies. From the coding process, four drivers for system improvement emerged: (1) system-wide approaches; (2) human capital; (3) governance; and (4) availability of resources. Each of these drivers highlight key explanations that are found in the literature. These drivers relate to social and political theories of education and express different views on the way state interventions interact with pedagogical practices at a systemic level. They are also influenced by the contexts in which each study was done, and the different interventions that states attempted. Nevertheless, the papers that we reviewed did not consider these drivers as isolated from each other. We are artificially separating them as an analytic strategy to map the overall different themes that are

mentioned in the literature. However, we will also try to account for the connections among the different drivers that were mentioned in our corpus.

Findings

System-wide approaches

Most of the texts that we analyzed stressed the importance of considering the approaches to system-wide improvement taken in different educational systems. In some cases, critical views of single reforms ("silver bullet" interventions) or single school improvement strategies were used to emphasize the need for a system-wide focus. A common limitation of improvement focused on specific schools is scale (Harris, 2010). The other problem is that focusing on improving individual schools "always distorts social equity" (Hopkins, 2007). Thus, alternative models that can deliver change on a larger scale are promoted in the literature. These involve changes at multiple dimensions and scales of education systems (Dinham et al., 2011; Harris, 2010).

What is needed then, as Elmore (2007) suggests based on his research on the Victoria case, is a strategic view, since "systemic problems require multidimensional, integrated, highly-focused, and coherent interventions, defined by a clear theory of action, in order to break the equilibrium of an underperforming system." Some authors even mention the need for more holistic reforms in social and economic policies that could improve the socio-economic milieu in which students and their families live (Hopkins et al., 2007).

An agenda for continuous improvement should be focused on outcomes, processes, and a system of continuous learning across different actors in the system (O'Day & Smith, 2016). For Fullan (2016), systems improvement should be based on three principles: "deep change in the culture of learning, local ownership of the learning agenda, and a system of continuous improvement and innovation that is simultaneously bottom-up, top-down and sideways."

Elmore (2007) notes that systemic approaches have to be complex and simple at the same time. On the one hand, complex because they have to impact simultaneously on social, organizational, technical, and cultural dimensions of the education system. On the other hand, systemic approaches need to be simple because they must be based on a straightforward narrative of change; systemic approaches need also be based on a simple message that all stakeholders can easily appropriate.

The importance of interventions at the level of values, beliefs, and expectations of actors within the system is one of the most common driver across the texts that we reviewed. Overall, most authors emphasize the need to define common goals and a coherent framework as a critical element of system-wide initiatives. They also highlight the need to develop trust and alignment of stakeholders around those goals.

The case of Ontario is frequently used as a paradigmatic case of a systemic improvement strategy that focused on a small number of ambitious goals (Fullan & Rincon-Gallardo, 2016). Levin (2012) suggests that a balance must be struck between ambitious but realistic and achievable goals. Sometimes organizations

tend to aim too low as a protective strategy, but aiming at impossible goals can also be detrimental. In his view, core goals should also be well-grounded and publicly stated.

Goals should be measurable in some way to track progress, but there are also some dangers associated with measuring the attainment of goals (Levin, 2012). One is that performance indicators could displace worthwhile goals. The aim is for students to develop, for example, meaningful literacy skills, not to learn to perform well in standardized tests mechanically. Another danger is that the focus on a small set of crucial goals can become too narrow. Other significant educational goals and activities, such as arts or physical education, can be left behind.

Once a small number of core goals is defined, the next challenge mentioned in the literature is the need for coherence and appropriation of these goals in the whole system. Fullan (2016) refers to "shared coherence," that according to the literature, has been vital in improvement strategies in Ontario (Campbell, 2020), Wales (Harris, 2010; Harris & Jones, 2017), New Zealand (Dinham et al., 2011) and Victoria in Australia (Elmore, 2007).

The continuity in time of core goals and frameworks agreed upon in the whole system is another critical aspect of successful systemic improvement strategies highlighted in the literature (Elmore & Burney, 1998; Rivas et al., 2020; Snipes et al., 2002). In some cases, such as Long Beach (Zavadsky, 2016) and Chile (Cox, 2004), political stability has helped maintain goals and frameworks, while McAleavy and Elwick (2016) suggest that the main aspects of the reforms they analyze in London continued even with a change in political control. As a counter-example of agreed, long-standing reforms, Betts et al. (2005) suggests that the reforms he analyzed in San Diego had generated intense controversy with teacher unions and other stakeholders. Thus, when the superintendent who championed the reform left the district, the reforms were substantially dismantled.

Another general principle highlighted in the literature as key in system-wide improvement efforts is the emphasis on equity and the most disadvantaged schools and/or students (Elmore, 2007; Fullan, 2016; O'Day & Smith, 2016; Snipes et al., 2002). New Zealand, for example, took a two-step approach to improvement. In the first one, between the 1990s and 2009, it addressed schools that catered for low-income families and had problems in governance or student achievement. The second phase aimed to improve the whole system's performance (Dinham et al., 2011). Similarly, the P.900 program in Chile implemented special measures to support learning results in the 10% of primary schools with the worst results (Cox, 2004).

A final component of successful systemic improvement approaches is the emphasis on instruction and student learning as the key elements that should drive all aspects of the reforms (O'Day and Smith, 2016; Snipes et al., 2002). In most cases, this component was strongly linked to the discussion of strategies to improve human capital within educational systems and with the pedagogical governance capacities of the state.

Human capital: The virtuous circle of improving teacher quality

The need to invest in human capital, especially teachers, was the most salient specific driver in the literature we explored. There were different views and approaches, some stressing the importance of promoting collaboration among teachers, others the importance of considering teachers' needs and expertise. Some stressed training, while others emphasized support mechanisms, including mentorship schemes. Most of the studies pointed to the relevance of helping teachers focus on instruction and student learning. Overall, capacity building (strongly related to system-wide approaches) and support for teachers focused on instruction (strongly related to governance and bridging macro policies and micro practices) were the salient themes that were mentioned as fundamental to promote system-wide improvement.

Again, the case of Ontario is paradigmatic in the literature. Osmond-Johnson and Campbell (2018) note that one of the main strategies for developing professional capacity is recruitment, training, development, recognition, working conditions, and career progression. They note that a critical element was the combination of capacity building and support for teachers focusing on results. Similarly, Elmore (2007) notes that the central message of the Victorian (Australia) model is that "schools improve by investing thoughtfully and coherently in the knowledge and skill of educators. Everything else is instrumental to this purpose." Pang and Miao (2017) also stressed the importance of developing teachers as leaders in Shanghai's reform. Harris and Jones (2017) suggest that what is fundamental in an improvement strategy in education is affecting classroom practice. They analyze the process through which, in 2008, the Welsh Government aimed at establishing strong professional learning communities in schools so that practitioners could share their professional knowledge to improve student outcomes.

Similarly, one of the critical elements of the strategy in District #2 in New York City was the creation of solid professional networks among teachers and principals across schools to reduce isolation among practitioners (Elmore & Burney, 1998). The Sanger Unified District in California also promoted collaboration and shared responsibility among teachers (David & Talbert, 2012).

Another critical characteristic of strategies to improve teachers as part of systemic reforms discussed in the literature is to promote contextualized professional development opportunities. For example, in New Zealand, Dinham et al. (2011) refer to "developing capacity at the point of delivery." Levin (2012) highlights the need for learning in context. Similarly, in London, school-based coaching replaced off-site workshops as the primary professional development format (McAleavy & Elwick, 2016).

The literature highlights that the aim of capacity building among teachers should be focused on improving instruction (Betts et al., 2005; O'Day & Quick, 2009; Zavadsky, 2016). Capacity building was an essential strategy for teachers and school leaders (Dinham et al., 2011; Levin, 2012). The importance of leadership in successful educational reform is well documented in the literature, stressing that good leadership must be developed and cultivated through recruitment and professional development strategies. An example of a critical aspect of leadership changes is reported for the case of the Sanger school district in California that "shifted district

and school administrator's thinking and skills from being managers to being leaders of learning" (David & Talbert, 2012). In San Diego, instructional leadership was also promoted (O'Day & Quick, 2009).

The notions of distributed and shared leadership were also emphasized, aiming at leaders that respect the professional knowledge and practice of those under their responsibility (Osmond-Johnson & Campbell, 2018). In London, coaching was also used to promote horizontal learning among headteachers and recognize exceptional leaders (Zavadsky, 2016). Similarly, in District # 2, New York City, a principal mentorship program was designed in which more- and less-experienced principals would work with each other (Elmore & Burney, 1998).

Governance and macro–micro level bridges

Changes in governance are also mentioned in the literature as relevant components of reforms aimed at system-wide improvement. Levin (2012) stresses the importance of paying attention to implementation processes. "Announcements followed by documents and a few training sessions will not change what happens in a large, complex system." Within the ability of the central authorities to lead and support the process of change, the tension between tight central prescription and control, and local and school autonomy is a permanent challenge that is mentioned in the literature (Campbell, 2020; Dinham et al., 2011; Snipes et al., 2002). Fullan (2016) suggests "shaping and cultivating a culture of purposeful learning that is neither tightly controlled nor too loose."

Following this direction, a recent comparative study of subnational educational improvement found that building a governance platform was the key element shared by the twelve cases of improvement among 486 systems in Latin America (Rivas et al., 2020). A governance platform involves reflective leadership, understanding the education system culture, creating long-lasting legitimacy and trust, and building structural and highly sophisticated channels to influence classroom practices. This conclusion is close to the findings of a major comparative study on districts improvement in the United States (Zavadasky, 2016).

Other studies also found that a scripted curriculum is critical in developing a systemic approach to improving disadvantaged contexts. In South Africa, a vital component of the improvement strategies was the distribution of scripted lesson plans, the provision of quality learning materials, and instructional coaching (Fleisch, 2016). Similarly, other studies that looked at the subnational level also acknowledged the importance curricular coherence and alignment as a central ingredient to achieve systemic improvement (Campbell, 2020; Elmore & Burney, 1998; O'Day & Quick, 2009).

The notion of alignment between different instructional components is key in classical studies of curriculum governance (Fuhrman, 1993; Porter, 2002). Well-aligned systems provide common expectations to teachers through a coherent "curricular environment" of instruments intended to steer teaching (Valverde, 2014).

This view shows the importance of bridging central government policies with real life pedagogical practices, forming an internal architecture of curricular coherence,

defined as “articulated over time as a sequence of topics and performances consistent with the logical and, if appropriate, hierarchical nature of the disciplinary content from which the subject matter derives” (Schmidt et al., 2005, p. 528).

In other studies, the key to systemic improvement was found in the political economy of reforms. In Ecuador, the alignment of political forces turned union resistances down to introduce higher standards for recruitment and regular evaluation of teacher performance (Schneider et al., 2019). This case shows the connections between governance mechanisms and teacher-related policies (a theme analyzed separately here that has tight mutual connections).

One of the areas in which the relation between the central authorities and local or school leaders is evident is accountability policies. Most of the texts we reviewed stressed the need for accountability measures, but not all agreed on the values that should underpin these policies. While some stress support mechanisms, considering that sanctions are detrimental to improvement, other stress that more punitive approaches are essential to make different stakeholders accountable for student learning and overall outcomes. However, even those that promote more high-stakes accountability policies tend to balance these with a participative definition of standards and the of use data about outcomes to develop support mechanisms, rather than purely for defining sanctions.

In Ontario, they refer to the principle of promoting "professional accountability: results without rancor or ranking” (Osmond-Johnson & Campbell, 2018). When underperformance is identified, non-punitive interventions are promoted, based on support mechanisms to help and enable improvement in student learning (Fullan & Rincon-Gallardo, 2016). In Victoria, Australia, accountability was aligned with setting priorities for human capital development (Elmore, 2007).

Elmore (2007) mentions the US as a negative example in which a relatively punitive, regulatory approach to accountability is producing many more low-performing schools. The most extreme case of a punitive approach that we found in the literature we explored was London. According to McAleavy and Elwick (2016), comparative data allows benchmarking as a critical component of the improvement process. Failing schools are "replaced with new schools, removed from district control and run instead by a government-approved ’sponsor’ in the form of a not-for-profit trust provided with philanthropic support by the sponsor.”

In all US school districts reported in the literature, accountability schemes tended toward more punitive approaches. However, they are combined with support mechanisms and other features to moderate some of their potential adverse effects (David & Talbert, 2012; Elmore & Burney, 1998; Zavadsky, 2016). In the case of Poland, the introduction of nationwide comparable exams, conducted at the end of every stage of education, played a crucial role in assuring quality in the education system (Jakubowski, 2015).

Some authors considered that a crucial element of governance is implementing evidence-based policies (Fullan & Rincon Gallardo, 2016; Levin, 2012; Snipes et al., 2002). This implies initially designing policies based on research that can provide a valid assessment of the problems that need to be addressed, as well as on research that can point to good practices and approaches that have worked in other places (Osmond-Johnson & Campbell, 2018; Snipes et al., 2002).

Research is also needed to collect data and learn from implementations. Levin (2012) stresses the need for a continuous learning strategy through research and data that allow the system to learn from its performance and make the necessary adjustments. Fullan and Rincon-Gallardo (2016) highlight the need to "foster learning from implementation laterally (across schools and districts), and vertically (between schools and districts)." Similarly, in Long Beach, research-based and targeted responses are considered a key component of improvement strategies, while Snipes et al. (2002) emphasize the capacity to diagnose instructional problems and design data-driven policies across all system levels. In Poland, the concept of 'knowledge mobilization' was used to explain the country's improvement in PISA (Fazlagić & Erkol, 2015).

Availability of resources

Several authors also mentioned the importance of effective allocation of resources aligned with core goals and improvement strategies to support improvement efforts (Levin, 2012; Osmond-Johnson & Campbell, 2018). In some cases, such as New Zealand and Chile, funding was funneled toward disadvantaged schools to promote equity (Cox, 2004; Dinham et al., 2011). The distribution of learning materials to improve learning was primarily mentioned in education systems in less wealthy countries, such as South Africa and Chile.

In South Africa, a vital driver of the improvement strategies was the provision of quality learning materials, and instructional coaching (Fleisch, 2016). This example shows the connections between curricular and pedagogical governance mechanisms through resource allocation. Two themes analyzed separately in this paper, but that are strongly connected. In Chile, public expenditure in education more than tripled between 1990 and 2003, and private expenditure followed a similar trend. The national state distributed textbooks in five subjects to all students in subsidized primary education (92% of the total). The state also provided computers and Internet access to primary and secondary education (Cox, 2004).

Another vital resource is time. Increasing the time students spend learning in schools is another strategy that a few authors mentioned in the texts we analyzed. In Chile, time in instruction was expanded across the system (Cox, 2004). In Poland, increased hours of instruction and delayed tracking of students into the vocational education stream were also considered essential factors in improving test scores (Bin Mahfooz & Hovde, 2010). In some cases, the extra time was an opportunity for students with low performance and greater academic need (Betts et al., 2005; Elmore & Burney, 1998).

Discussion: Emerging lessons from systems' improvement

Our review of educational research aimed at understanding the kind of interventions that can contribute to the improvement of educational systems. The systematic literature review has taken us through a journey across methodological

approaches and very different educational contexts. We have also travelled in time, as we identified three historical stages in the study of system-wide improvement in education. The first is the broad tradition of comparative education research focused on "notable cases" that mostly used unsystematic direct observation and, later, some data on access to educational institutions to define that one system was better than another. In this stage the focus was on spatial comparisons that contrasted different educational systems, rather than on comparisons across time that could point to improvements in one system. The second stage we identified focused on educational districts within the United States, as the rise of standardized assessments in this country made intertemporal comparisons possible (Elmore & Burney, 1998; Murphy & Hallinger, 1988; Snipes et al., 2002; Wenger et al., 2002). The third stage is the post-ISLA (especially PISA) era of international comparisons of systems´ improvement, with the notable influence of the "McKinsey" report and the multiple studies reviewed in this paper.

Within this third phase of research, we found studies that had very different methodological approaches and looked at a variety of educational contexts. Most of the empirical papers that we included in this review used an either quantitative or a qualitative research design; only some of them followed a mixed-methods approach (David & Talbert, 2012; McAleavy & Elwick, 2016; O'Day & Quick, 2009; Rivas et al., 2020). The latter were able to offer a more convincing story concerning why the systems improved. The use of both qualitative and quantitative methodologies allowed the authors to offer a more complete picture to explain the multidimensionality of a system-wide improvement.

We should also acknowledge that in these mixed methods studies, the researchers looked at subnational systems. It is interesting to think if the difference in level (i.e., from national to subnational) helps the researcher bound the study and focus on more particularized explanatory drivers of the improvement of the educational system. Thus, the question that still remains is whether the mixed-methods and comparative approaches studies were more effective in coming up with solid evidence to support the theory of change because of the methodology that the researchers used, or because of the scale of the study.

Nevertheless, within these studies we could identify some common patterns and drivers in terms of the questions they were addressing, as they all aimed at understanding what are the kind of policy interventions that could explain system wide improvements.

Most of the empirical research on concrete education systems that managed to improve over time in learning assessments find multiple factors as concurrent explanations. There are very few studies that found a "silver bullet approach" (Jakubowski, 2015). Instead, we found different emphasis or what we call "drivers" as a key concept to differentiate the literature.

Our review shows that the need to design governance mechanisms that could bridge macro curricular decisions with everyday pedagogical practices is a fundamental challenge for the improvement of educational systems. The notion of alignment between different instructional components is key in classical studies of curriculum governance (Fuhrman, 1993; Porter, 2002). The need for this alignment also relates to the importance of having a clear systemic strategy based

on measurable outputs that can be appropriated by actors at different levels of the educational systems. Well-aligned systems provide common expectations to teachers through a coherent "curricular environment" of instruments intended to steer teaching and learning (Valverde, 2014). In some of the studies we reviewed, this curricular coherence was sought through the distribution of resources aligned with core goals, fundamentally student cognitive achievement.

What is also clear from this review is that attaining system-wide improvement is a complex endeavor that requires a variety of strategies and approaches that cannot be construed as a simple “silver-bullet” recipe. On the contrary, we have systematized a set of drivers that have been used in different parts of the world. These can be useful as part of a conceptual framework to open up contextualized conversations among different stakeholders to find specific and localized strategies that would probably have to be explored, experimented and adjusted in the process.

There were also certain commonalities in that most of these studies linked, in slightly different ways, the notion of improvement with results from large scale standardized student learning assessments. Logically, this focus on learning assessments contributed to an emphasis on interventions that promote student learning, rather than on strategies that could have other aims, such as promoting access to more years of schooling or different curricular perspectives.

The performative turn in educational systems has made student learning the holy grail of education policies (Rizvi & Beech, 2017). The quest to influence learning permeated all the other factors that we identified in the literature. So, for example, the importance of promoting teacher quality stressed the need to focus professional development efforts on instruction and student learning and tended to prefer contextualized training strategies that affected classroom practice. Similarly, when the challenge of governance mechanisms to promote improvement were discussed, the strategies to connect macro decisions with student learning were salient. Furthermore, debates on the approach to accountability policies that should be pursued (whether centered on more control or in trust and autonomy) were framed in terms of different ideological stand points. These two approaches could contribute to manage the work on teachers so that they would embrace the responsibility for student learning.

We believe that even if no straightforward recipe can be devised, research on the kinds of interventions that can promote system-wide improvement of educational systems is a worthwhile endeavor. The variety of theoretical and methodological approaches that we found is a valuable sign of how vibrant the theme can be, both from an academic and a political perspective.

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